

AMENDMENT NO. 1 MARCH 1979
TO
IS : 8101-1976 SPECIFICATION FOR
POSTER COLOURS

Alteration

(*Pages 5 and 6, Appendix A*) — Substitute the following for the existing appendix:

'APPENDIX A
(*Clause 3.2*)

DETERMINATION OF LEAD

A-1. APPARATUS

A-1.1 Separating Funnel — of 500-ml capacity.

A-1.2 Nessler Tube — of 50-ml capacity.

A-2. REAGENTS

A-2.1 Dilute Nitric Acid — lead-free, approximately 4 N.

A-2.2 Ammonium Acetate Solution — lead-free, 10 percent (*m/v*).

A-2.3 Citric Acid Solution — lead-free, 0.5 g per ml.

A-2.4 Ammonia — Concentrated, of redistilled quality.

A-2.5 Potassium Cyanide Solution — 10 percent (*m/v*).

A-2.6 Standard Lead Solution — Dissolve 0.40 g of lead nitrate [$\text{Pb} (\text{NO}_3)_2$] in water containing 2 or 3 ml of concentrated nitric acid and make up the volume to 1 000 ml with water. Transfer 10 ml of this solution to a volumetric flask, add 2 or 3 ml of concentrated nitric acid and dilute with water to 1 000 ml. One millilitre of this solution contains 2.5 μg of lead (as Pb). The diluted solution shall be freshly prepared.

A-2.7 Dithizone Solution — Dissolve 0.01 g of dithizone in 100 ml of carbon tetrachloride, shaking intermittently for 1 hour. Allow to stand overnight and shake once again before using. This shall be kept in a cool and dark place. This gives a 0 01 percent solution. Filter, if necessary.

Dilute 10 ml of this solution to 100 ml with carbon tetrachloride in a 100-ml volumetric flask. This shall be prepared fresh before determination. This gives a solution of 0.001 percent.

NOTE 1 — Carbon tetrachloride used should be further purified. One litre of carbon tetrachloride is extracted with two portions of 25 ml of dilute ammonium hydroxide and then kept over 100 g of activated carbon. Before use, it is decanted and distilled at about 80°C over a little fresh lime.

NOTE 2 — Sometimes dithizone solid and its 0.01 percent solution deteriorate on storage. The 0.01 percent solution should, therefore, be tested before further dilution, by shaking 2 ml of the solution with 5 ml of 1 percent ammonium hydroxide. If the organic layer is only faintly yellow under these conditions the solution may be used. If it is deeply coloured it shall be discarded and fresh solution be prepared. The solution as well as the reagent should be stored in a refrigerator and exposure to sunlight should be avoided during analytical work. To increase the stability of 0.01 percent solution, it should be covered with a thin aqueous layer saturated with sulphur dioxide.

A-3. PROCEDURE

A-3.1 Treat about 20 g of the material accurately weighed, with dilute nitric acid in a porcelain basin. The quantity of the acid used is immaterial provided it is sufficient to extract the soluble matter; but avoid too great an excess since it has to be evaporated off. Allow the basin to stand on a boiling water-bath for at least 3 hours. In case a large quantity of insoluble residue is left, heat the basin on a water-bath overnight. Decant off the supernatant liquid through a quantitative filter paper and extract the insoluble residue again on a boiling water-bath for one hour with dilute nitric acid. Filter through the same filter paper and wash the residue on the filter paper thoroughly with hot water*. Collect the filtrates and washings in a 500-ml separating funnel. Add about 2 ml of citric acid and make the solution ammoniacal with pH between 8.5 to 10 (bluish-green to blue towards a drop of thymol blue), after neutralizing and adding 5 ml of 10 percent potassium cyanide solution. Extract the lead with successive 20 ml portions of dithizone reagent. Drain the organic layers into a Nessler tube and stir well.

Take several aliquots of the standard lead-solution into a series of separating funnels, and develop the colour as in the case of sample. Extract the organic coloured layers into a series of Nessler tubes. Compare the colour developed from the sample solution with those of the standard solutions. Note the volume of the standard with which the colour of the test solution matches. If the colour of the test solution is intermediate between two standard solutions then repeat the experiment by taking more number of standard solutions in that range and arrive at exact colour matching.

*Treat the residue on the filter paper with 10 ml of ammonium acetate solution, filter and wash with hot water.

A-4. CALCULATION

$$\text{A-4.1 Lead (as Pb), ppm} = \frac{V \times f}{M} \times 10^6$$

where

V = volume in ml of standard lead solution matching with the test solution,

f = mass in g of lead equivalent to 1 ml of standard lead solution, and

M = mass in g of the material taken for the test.'

(CDC 13)